INCH-POUND

MIL-E-22200/4D

29 August 1991

SUPERSEDING
MIL-E-22200/4C
31 October 1979

MILITARY SPECIFICATION

ELECTRODES, WELDING, COVERED, COPPER-NICKEL ALLOY

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 70:30 copper-nickel covered electrodes used for fabrication and repair welding of copper-nickel alloys.

1.2 Classification. The electrodes shall be furnished in the type, classes, and sizes shown in table I, as specified (see 6.2):

TABLE I. *Types, classes, and sizes.*

Type	Class	Positions of welding	Sizes (inch)
MIL-CuNi (70:30)	1	All	3/32, 1/8
	2	Horizontal fillets and flat	5/32, 3/16

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3439

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-C-15726	Copper-Nickel Alloy Rod, Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings
MIL-E-22200	Electrodes, Welding, Covered; General Specification for

STANDARDS

MILITARY

MIL-STD-271	Requirements for Nondestructive Testing Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

NAVSEA 0900-LP-003-9000 Radiographic Standards for Production and Repair Welds.

(Copies of publications are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 122	Standard Specification for Copper-Nickel-Tin Alloy, Copper-Nickel-Zinc Alloy (Nickel Silver), and Copper-Nickel Alloy Plate, Sheet, Strip, and Rolled Bar **(DOD Adopted)**

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

AMERICAN WELDING SOCIETY (AWS)

B4.0	Standard Methods for Mechanical Testing of Welds **(DOD Adopted)**

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33135.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. Requirements

3.1 Electrodes furnished under this specification shall conform to the requirements of MIL-E-22200 and as specified herein.

3.2 **Coverings.**

3.2.1 **Composition.** The chemical composition of the electrode covering material is optional with the manufacturer, except moisture content, subject to material requirements of MIL-E-22200. The total water content in the covering shall not exceed 0.8 percent by weight.

3.3 **Mechanical properties.**

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3.3.1 **Tensile test.** All weld tensile samples shall have a minimum tensile strength of 50,000 pounds per square inch (lb/in^2) and a minimum elongation of 30 percent in a 4x diameter gauge length.

3.3.2 **Bending.** After bending, the weld metal of each specimen shall be visually examined on the convex surface by the unaided eye and shall not exhibit evidence of fissuring in excess of two fissures each having a maximum length of 1/8-inch nor exhibit evidence of any fissuring over 1/8-inch long measured in any direction. Cracks occurring on the corners of a specimen during testing shall not be considered.

3.4 Chemical composition. The chemical composition of deposited weld metal shall be as specified in table II.

TABLE II. *Chemical composition of deposited weld metal.*

Element[1]	Percent[2]
Copper	Remainder
Zinc	*
Tin	*
Manganese	1.00 – 2.50
Iron	0.40 – 0.75
Silicon	0.50
Nickel(including cobalt)	29.0 – 33.0
Lead	0.01*
Titanium	0.50
Phosphorus	0.020
Sulfur	0.015
Total other elements[3]	0.50

[1]Analysis shall be made for the elements shown in this table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine that the total of these other elements is not present in excess of the limits specified for "total other elements" in the table. Specific values shall be reported for the results of each element analyzed including other elements when applicable.

[2] Single values shown are maximum percentages, except where otherwise specified.

[3]Total other elements, including elements marked with an asterisk (*) shall not exceed the value specified.

3.5 Electrode identification. Electrodes shall be identified as specified in MIL-E-22200.

3.5.1 **Heat and lot identification.** When specified, each electrode shall be marked with heat or lot identification as specified in the marking requirements of MIL-E-22200 (see 6.2).

3.6 Nominal lengths. The specified nominal lengths of electrodes shall be in accordance with table III.

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TABLE III. *Nominal lengths.*

Sizes (inch)	Electrode lengths end grip type (inches)
3/32	9 or 12
1/8, 5/32, 3/16	14

3.7 Core wire alloy identity. The core wire alloy identity requirements shall be as specified in MIL-E-222001.

3.8 Groove weld soundness. The soundness of groove welds shall meet the requirements of grade I of NAVSEA 0900-LP-003-9000 as specified in the soundness requirements of MIL-E-22200.

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-E-22200 and as specified herein.

4.2 Qualification tests. Qualification tests shall be as specified in MIL-E-22200 and tables IV and V herein.

4.3 Quality conformance inspection.

4.3.1 Lot. A lot of electrodes shall be as specified in MIL-E-22200 for inspection level A.

4.3.2 Sampling for quality conformance inspection. Sampling of containers and electrodes for quality conformance inspection shall be in accordance with the sampling requirements of MIL-E-22200.

4.3.3 Quality conformance inspection. The samples selected in accordance with 4.3.2 shall be subjected to the tests specified in tables IV and V.

4.3.4 Unsatisfactory test results. If the results of the first tests are determined to be unsatisfactory, two retests of each failed test shall be required. The results of both retests shall be satisfactory for quality conformance inspection.

4.4 Test procedures.

4.4.1 Weld tests. Table IV summarizes the weld tests required under this specification for qualification testing of electrodes for the Qualified Products List (QPL) and for quality conformance inspection.

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TABLE IV. *Summary of required weld tests.*

Test schedule	Diameter sizes				Weld joint design procedure[1]	Welding position[2]	Tests				Requirements
	3/32	1/8	5/32	3/16			Radio-graphic	Tensile	Bend	Chemical analysis[3]	
A	X	X	X	X	Figure 2 and notes	F	–[4]	X	X	X	MIL-E-22200 3.3 and table II herein
B	X –	X –	– X	– X	Figure 3 and notes	V F	X X[4]	– –	– –	– –	MIL-E-22200

[1]Direct current, reverse polarity shall be used for all weld tests, with amperages and welding techniques recommended by the manufacturer. Electrical conditions shall be recorded.

[2]F = flat, V = vertical.

[3]May be determined from drillings taken from the weld or from the ruptured ends of the tensile specimens. In the event of variances with composition requirements or conflicting results by testing activities, analysis shall be verified by procedure specified in MIL-E-22200.

[4]When both test schedules (A and B) are applicable, radiography of test A will satisfy the requirements of test B for class 2 electrodes, if test A is radiographed, test B may be eliminated for class 2 electrodes.

4.4.2 Other tests. Table V summarizes tests, in addition to those specified in 4.4.1, required for qualification testing of electrodes for the QPL and quality conformance inspection as applicable.

TABLE V. *Summary of other tests.*

Tests	Procedures	Requirements
Tolerance on diameter and length of core wire	–	MIL-E-22200 and 3.6 herein
Concentricity	–	MIL-E-22200
Dielectric strength[1]	MIL-E-22200	MIL-E-22200
Covering, flaking and cracking[1]	–	MIL-E-22200
Covering moisture	–	3.2.1 herein
Alloy identity – core wire	–	MIL-E-22200

[1]Required for qualification inspection only.

4.5 Packaging inspection. The packaging, packing, and marking shall be inspected for compliance with section 5 herein.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified in MIL-E-22200.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type MIL-CuNi (70:30) covered electrode is intended for use in shielded metal-arc welding of copper-nickel alloy tubing of MIL-T-16420, or plates and shapes of MIL-C-15726.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

c. Type, class, and size (see 1.2)

d. Whether heat or lot identification is required for individual electrodes (see 3.5.1)

e. For 3/32 size electrode, the length required (see 3.6 and table III)

f. Electrodes should be ordered by the pound.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
6.4	DI-MISC-80678	Certification data/report	–

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Certificate of test. When required by the contract or order, a certificate of quality conformance of the weld tests specified in 4.3.3 containing actual quantitive test and inspection results including the range of amperage employed in making the required welds for quality

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conformance inspection for each accepted lot and heat, as applicable, included in a particular shipment shall be furnished with the shipment. The certification shall include test results from the quality conformance test for the shipment. The minimum quality conformance test result data required shall be as listed on figure 1 and may be submitted on the form as shown (see 6.3).

6.5 Inspecton after delivery. Post delivery inspection of electrodes to determine conformance to this specification and for acceptance thereof is the responsibility of the consignee.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Subject term (key word) listing.

Alloy
Composition
Groove weld
Tensile strength
Mechanical properties

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - MR, AT

User activities:
Navy - OS, MC

Preparing activity:
Navy - SH
(Project 3439-0681)

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Manufacturer or Distributor _____

Address _____

Date _____

Specification MIL- _____

Type MIL- _____

Diameter & Length _____

Inspection Level _____

Lot No. _____

Chemical Analysis
(Complete)

Copper _____ Nickel (including Cobalt) _____

Zinc _____

Tin _____ Lead _____

Manganese _____ Titanium _____

Iron _____ Phosphorus _____

Silicon _____ Sulphur _____

Total other elements _____

Chemistry was taken from: [ ] Chem Pad
[ ] Groove Weld

Radiograph Results _____

Concentricity (percent) _____

Covering moisture (percent) _____

Alloy identity tests:

(a) Core wire _____

Customer's Name _____

Customer's Order No. _____

Core Wire Heat No. _____

Lot Identification MIL-E-22200
Paragraphs _____

Mechanical Test

As-Welded

Yield Strength
(0.2 percent offset method) _____

Tensile Strength _____

Elongation (percent in ____ inches) _____

Reduction of Area (percent) _____

Bend Test

Side Bend _____
Side Bend _____
Side Bend _____

Groove Weld Test

Test Schedule	A	B	Chem Pad
Amperage	_____	_____	_____

Grinding during Test Schedule B Plate Preparation (See Note 3) Operator Error (Layer(s))

_____

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

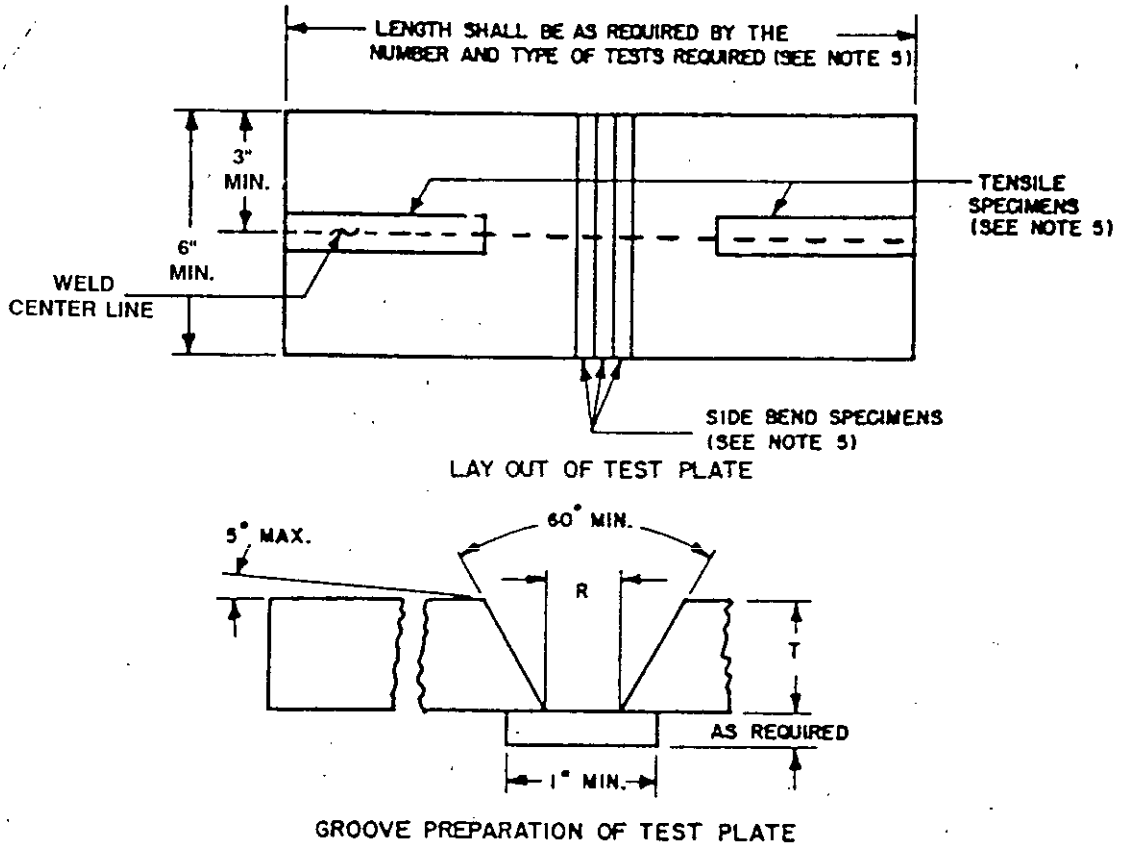
_____

NOTES:

1. Items not applicable to the type electrode involved or exempted by the acquisition document shall be marked N/A.
2. This form is applicable to the quality conformance tests required for each lot. A copy of the quality conformance test certification for the lot (figure 1) shall be forwarded with each shipment regardless of the identification of the wet mixes of covering mixture represented in this shipment.
3. See requirements for slag removal and grinding of MIL-E-22200.

FIGURE 1. *Sample certification of quality conformance tests.*

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Electrodes size (inch)	Minimum thickness (T) (inch)	Minimum root opening (R) (inch)	Number of layers[1]	
			Minimum	Maximum
3/32	1/2	1/4	_[1]	_[1]
1/8	1/2	1/4	_[1]	_[1]
5/32	3/4	1/2	6	9
3/16	3/4	1/2	6	9

[1]Pass and layer sequence shall be recorded and reported.

FIGURE 2. *Welded joint for all-weld-metal tension test and bend test (test schedule A).*

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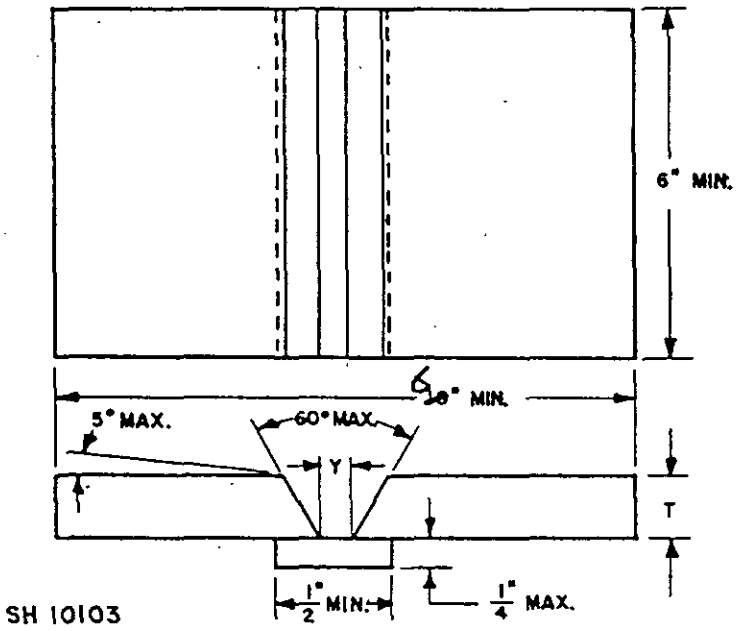
Notes to figure 2:

1. Welding shall be performed with plate in the flat position.
2. Base metal shall be in accordance with MIL-C-15726 or ASTM B122 (UNS No. C71500).
3. Preheat temperature shall be 60 °F minimum. The interpass temperature shall not exceed 300 °F.
4. Welds shall be deposited in approximately 1/8 inch thick layers using a 3 core wire diameter maximum weave build up sequence. Final layers shall include a weld reinforcement of 1/32 to 1/8 inch. The root weld layer may be deposited with 3/32 or 1/8 inch diameter electrodes when testing 3/16 inch diameter electrodes.
5. The number and type of mechanical tests shall be as follows:

Electrode type	Size (inch)	Number of Test specimens		
		Tensile		Sidebend
		0.250	0.505	
MIL-CuNi (70:30)	3/32	2		3
	1/8	2		3
	5/32		2	3
	3/16		2	3

6. Tensile and bend specimens shall be prepared and tested in accordance with AWS B4.0.
7. The results of tensile and bend tests shall conform to the requirements of 3.3.
8. Radiographic examination of test A is not required but may be used in lieu of radiography of test B for class 2 electrodes. When test A is employed for radiography, after completion of welding, the weld reinforcement and backing strip shall be removed flush with base plate surfaces. The assembly shall be radiographed in accordance with MIL-STD-271. Radiographs shall meet the acceptance standard of NAVSEA 0900-LP-003-9000 as specified for soundness in MIL-E-22200.

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Electrode size (inch)	3/32	1/8	5/32	3/16
Thickness (T) min. (inch)	1/4	3/8	1/2	1/2
Root opening (Y) (inch)[1]	1/4	5/16	3/8	1/2

[1]Tolerance ± 1/16 inch.

Notes to figure 3:

1. Welding shall be accomplished in the vertical or flat position as specified in table IV.
2. Base metal shall be in accordance with MIL-C-15726 or ASTM B122 (UNS No. C715000).
3. The bevel edge preparation shall be machined or ground.
4. Welds shall be deposited in approximately 1/8 inch thick layers. Each weld bead shall contain a start in the area to be evaluated. Weld deposits shall be made using a 3 core wire diameter maximum weld weave. The root weld layer may be deposited with 3/32 or 1/8 inch diameter electrodes when testing 3/16 inch diameter electrodes.
5. The completed weld reinforcement shall be 3/16 inch maximum.
6. Preheat temperature shall be 60 °F minimum. The interpass temperature shall not exceed 300 °F.
7. After completion of the weld, the weld reinforcement and backing strip shall be removed flush with base plate surfaces and the assembly shall be radiographed in accordance with MIL-STD-271.
8. The radiographs shall meet the acceptance standards of NAVSEA 0900-LP-003-9000 as specified for soundness in MIL-E-22200.

FIGURE 3. *Groove weld usability test (test schedule B).*

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-E-22200/4D	91-08-29

3. DOCUMENT TITLE

ELECTRODES, WELDING, COVERED, COPPER-NICKEL ALLOY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
Technical Point of Contact (TPOC): Mr. John Dorn (SEA 5142) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS	(1) Commercial 602-0205 TPOC: 703-602-0205 (2) AUTOVON 8-332-0205
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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